AMD Token Factory — Documentation

Overview

Token Factory is the model API catalog of AMD Radeon Cloud. It gives developers free, OpenAI- and Anthropic-compatible LLM endpoints running on AMD GPUs, plus one-click dedicated model deployments paid for with credits.

The catalog:

	Public Free Model APIs
What it is	Shared endpoints, always on, run by AMD
Cost	Free (a daily usage cap applies)
Models	AMD chooses them, and the list changes over time
Base URL	`https://developer.amd.com.cn/radeon/api/v1`
Endpoints	Chat completions, Messages (Anthropic), models list, token counting, OCR
Limits	Per-key rate, concurrency and spend caps

One API key (format `rc-` + 48 hex characters) works for every shared model. Any OpenAI client (curl, the openai SDK, LangChain, Cherry Studio) works by changing the base URL and key. Anthropic clients such as Claude Code also work through `/v1/messages`.

Getting started — redeeming your team API key

Your team has been pre-provisioned with an API key for this event. To reveal it, **every member** of your team must sign in and request redemption. The key is only shown once all members have redeemed.

1. **Sign in.** Open notebooks.amd.com/tokenfactory. You will be redirected to the AMD SSO login page. Sign in with your AMD AI Developer Program account (the same email your team was registered with).
2. **Check your team dashboard.** After sign-in you will see your team name, your email, the team size, and a list of all team members with their redemption status (redeemed or waiting).
3. **Click "Redeem API Key".** This marks your email as redeemed. If other team members have not redeemed yet, the page shows who is still pending.
4. **Wait for all members.** The API key is **not revealed** until every team member has signed in and clicked "Redeem API Key". Coordinate with your teammates to ensure everyone completes this step. Click **"Check Status"** to refresh the redemption list.

5. **Copy your API key.** Once all members have redeemed, the key and this documentation are displayed. Click "Copy to clipboard" to copy the key.

> **Important — shared rate limits.** Your entire team shares a single API key. All rate limits — RPM (Requests Per Minute), concurrent request cap, and the daily spend cap — are enforced **per key, not per person**. That means every call from every teammate counts toward the same limits (see the Rate limits table below). Coordinate with your team: avoid burst-sending requests at the same time, stagger your calls, and use exponential backoff on 429 responses. If one member saturates the RPM limit, everyone else on the team gets throttled too.

1. **Make a call.**

```
curl https://developer.amd.com.cn/radeon/api/v1/chat/completions \
  -H "Authorization: Bearer $RADEON_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{"model":"DeepSeek-V4-Flash","messages":[{"role":"user","content":"Say hello in one sentence
```

From Python with the OpenAI SDK:

```
from openai import OpenAI

client = OpenAI(base_url="https://developer.amd.com.cn/radeon/api/v1", api_key="rc-...")
resp = client.chat.completions.create(
    model="DeepSeek-V4-Flash",
    messages=[{"role": "user", "content": "Say hello in one sentence."}],
)
print(resp.choices[0].message.content)
```

From the Anthropic SDK, set `base_url` to `https://developer.amd.com.cn/radeon/api`, without `/v1`, because the SDK appends `/v1/messages` itself.

1. **Check your remaining usage.** Every API response includes `X-RateLimit-*` headers showing your remaining requests and spend. You can also check explicitly:

```
curl https://developer.amd.com.cn/radeon/api/v1/usage?include_recent=true \
  -H "Authorization: Bearer $RADEON_API_KEY"
```

This returns your total quota, used amount, remaining amount, reset time, RPM limit, and recent request details. Since the whole team shares one key, this reflects **combined** usage across all teammates.

Account review. Some new accounts are held for verification. While an account is in review, it is not issued an API key and cannot launch instances. If your key is missing, check your account's review status.

Models and pricing

All 10 shared models are free and marked experimental. The usage figures below are in points per 1M tokens and only show relative usage: "Free to use. Points show relative usage—not a charge." They count against a daily spend cap, which the API reports in USD.

Public Free Model APIs

Model ID	Publisher	Type	Context (tokens)	Input	Thinks by default	Input / Output / Cache read (pts per 1M)
DeepSeek-V4-Flash (weights: V4-Flash-0731)	DeepSeek	LLM	1,048,576	Text	No	0.14 / 0.28 / 0.0028
DeepSeek-V4.1-Flash	DeepSeek	VLM	1,048,576	Text, image	No	0.14 / 0.28 / 0.0028
DeepSeek-V4-Flash-Vision-Exp	DeepSeek	VLM	1,048,576	Text, image	No	0.14 / 0.28 / 0.0028
MiMo-V2.6-Flash	Xiaomi	VLM	1,048,576	Text, image, audio	Yes	0.14 / 0.28 / 0.0028
GLM-5.3-Flash	Z.ai	LLM	262,144	Text	Yes	0.15 / 0.50 / 0.03
Qwen3.8-Flash-Next	Qwen	VLM	262,144	Text, image	Yes	0.15 / 0.47 / 0.016
Qwen3.8-27B	Qwen	VLM	131,072	Text, image	Yes	0.50 / 3.00 / 0.05
MiniCPM5-2B	OpenBMB	LLM	131,072	Text	No (no thinking)	0.124 / 0.7425 / 0.124
Gemma-4-31B-IT (experimental)	Google	VLM	262,144	Text, image	No	0.50 / 3.00 / 0.05
MinerU2.5-Pro	Radeon Cloud	Document OCR	n/a	PDF or image	n/a	"Limited Free", billed per page

Every chat model supports streaming, tool calling (`tools`, `tool_choice`) and `response_format: {"type":"json_object"}`. MiMo-V2.6-Flash is the only one that also accepts strict `json_schema` and audio input. MinerU2.5-Pro uses its own endpoint, `POST /v1/ocr`.

The catalog changes as models are added and retired, so call `GET /v1/models` at runtime instead of hard-coding names.

API reference

The shared endpoints sit behind a gateway. The gateway validates each request against a fixed set of fields and silently drops anything else. Every path also works under `/api/v1/...`.

Endpoints (base URL `https://developer.amd.com.cn/radeon/api`)

Method and path	Purpose	Auth
`POST /v1/chat/completions`	OpenAI-compatible chat (the main endpoint)	`Authorization: Bearer rc-...`
`POST /v1/messages`	Anthropic Messages API, for Claude Code and the anthropic SDK	Bearer, or `x-api-key` + `anthropic-version: 2023-06-01`
`POST /v1/messages/count_tokens`	Token count estimate (for sizing only, not billing)	Bearer or `x-api-key`
`GET /v1/models`	Catalog with pricing, context length and capability flags	Bearer or session
`POST /v1/ocr`	MinerU2.5-Pro: a PDF or image goes in and Markdown per page comes out	Bearer

The shared endpoints do not serve legacy completions, embeddings, image, audio, rerank or `/v1/responses`. Those paths return 404.

Chat completions parameters

- **Accepted:** `model`, `messages`, `stream`, `temperature`, `top_p`, `max_tokens`, `presence_penalty`, `frequency_penalty`, `response_format`, `tools`, `tool_choice`, `parallel_tool_calls`, `reasoning_effort` (or `reasoning.effort`).
- **Silently dropped:** `stop`, `seed`, `logit_bias`, `logprobs`, `top_logprobs`, `top_k`, `min_p`, `repetition_penalty`. If you need these, use a dedicated endpoint.
- **Rejected with 400:** `thinking: {...}` and `reasoning.enabled`. Use `reasoning_effort` instead. On `/v1/messages`, use `output_config: {"effort": "high"}`.

- **Timeout:** 10 minutes for a non-streaming request. When streaming, the 10 minutes is the maximum gap between chunks. Streams end with `data: [DONE]`.

Thinking (reasoning) controls

Thinking text comes back in `choices[0].message.reasoning`, and its token count in `usage.completion_tokens_details.reasoning_tokens` (except Qwen3.8-27B). Thinking tokens count toward `max_tokens`, so a low limit can leave `content` empty.

Model	Accepted `reasoning_effort`	If omitted	`system` message rules
DeepSeek-V4-Flash, V4.1-Flash, V4-Flash-Vision-Exp	none, minimal, low, medium, high, xhigh, max	No thinking	Any position, several allowed; `system` or `developer`
MiMo-V2.6-Flash	none + usual tiers (none turns it off)	Thinks	Any position; `developer` accepted
Qwen3.8-Flash-Next	none, low, medium, xhigh	Thinks (xhigh)	Exactly one, first; `system` only
Qwen3.8-27B	low, medium, xhigh (`high` = 400)	Thinks (xhigh)	At most one, first
GLM-5.3-Flash	low, medium, high (`xhigh` = 422)	Thinks	Any position; `system` only (`developer` = 422)
MiniCPM5-2B	Not applicable	No thinking	Any position; `system`
Gemma-4-31B-IT	On / Off only	No thinking	Use `system` as the first message

One request shape that works on every chat model: a single `system` message at index 0, `reasoning_effort` of `low` or `medium` (or omitted), and text-only content unless the model takes images. Size `max_tokens` to the model's context window.

Image, audio and OCR input

- **Images:** add an `{"type":"image_url","image_url":{"url":"..."}}` part to `content`. This works on DeepSeek-V4.1-Flash, V4-Flash-Vision-Exp, MiMo-V2.6-Flash and both Qwen3.8 models. GLM-5.3-Flash rejects images.
- **Audio (MiMo-V2.6-Flash only):** `{"type":"input_audio","input_audio":{"data":"<base64>","format":"wav"}}`.
- **OCR:** send `{"model":"MinerU2.5-Pro","document":{"type":"document_url","document_url":"https://..."},"pages":"0-2"}` to `/v1/ocr`. The response returns `pages[].markdown` and `usage_info.pages_processed`. It does not stream, so set a long client timeout.

Rate limits (shared endpoints)

Limit	Scope	Typical value
Requests per minute	Per API key (platform)	30
Requests per minute	Per account (gateway, 60 s sliding window)	20
Requests per minute	Per IP	120
Concurrent requests	Per API key	8
Spend cap	Per account, rolling daily period	Read it from `/v1/usage`

Every response carries `X-RateLimit-*` headers with your remaining requests and spend. When you are limited, you get a 429 with `Retry-After` (60 s for per-minute limits, 1 s for concurrency limits). Retry only 429s, with exponential backoff and jitter. Dedicated endpoints have none of these limits.

Usage and account endpoints

These Platform API paths use the same host as the model API. They accept a Bearer key.

- `GET /v1/usage?include_recent=true`: total quota, used amount, remaining, reset time, RPM limit and recent requests.
- `POST /api/profile/api-token`: rotate your key. The old key stops working immediately.
- `POST /api/credits/redeem` with `{"coupon": "..."}`: add credits.
- Instances: `POST /api/notebook/request`, `GET /api/notebook/status`, `DELETE /api/notebook/current`. Templates: `/api/profile/templates`.

Errors

Status	Meaning
400	Bad parameter, unknown model (`Requested model <name> not supported`), or not enough credits
401	Missing or invalid key
403	Account still in review (`account_not_verified`), or not your instance
404	The path is not served
422	Value rejected while the body was being parsed (for example `xhigh` on GLM-5.3-Flash)
429	Rate limited. Honour `Retry-After`
502 / 503	Gateway unreachable or saturated. Retry with backoff

Gateway errors use the OpenAI shape (`{"error":{...}}`). Errors from the platform layer are wrapped in `{"detail":{"error":{...}}}`, so check both.

FAQ, terms and support

FAQ

- **Do I need an instance to use the models?** No. Shared endpoints are free and always on. You only need an instance for a shell, a notebook, or a model the catalog doesn't offer.
- **Can I share my key or endpoint with a teammate?** For event/hackathon teams using Token Factory, yes — each team shares one API key and the shared rate limits apply to the whole team. Outside of event-provisioned keys, the general Radeon Cloud policy is one key per person. A dedicated endpoint only accepts its owner's key.
- **Are embeddings available?** Not on the shared endpoints. A vLLM dedicated endpoint usually serves `/v1/embeddings`.
- **A model that used to work now returns 400 "not supported".** The catalog changed. Re-check `GET /v1/models`.
- **My response has empty `content` and `finish_reason: length`.** Thinking used up `max_tokens`. Raise the limit, or send `reasoning_effort: "low"` (or `"none"` on models that allow it).
- **Does a dedicated endpoint URL stay the same?** No. Each launch creates a new instance ID and a new base URL.

API terms of use

- **Trial and development only.** The service has no SLA, uptime commitment or support obligation, and models can change or be withdrawn with little notice. Do not run production workloads on it.
- **Upstream models, unmodified.** The models are served from the original published weights, and each model's own licence applies. AMD gives no warranty for outputs.
- **Free.** Credits, quotas and points have no monetary value.
- **Shared team key.** Each event/hackathon team is provisioned one API key shared by all team members. Keys are disabled for selling, automated sign-ups, and use by multiple accounts outside the registered team.
- **No reselling.** You may not proxy, resell or re-badge the API, or tell a model to misreport which model it is.
- **Acceptable use.** No illegal content, malware, attacks, bypassing limits, CSAM, harassment or impersonation. Do not send sensitive personal, health or financial data.
- **Logging.** Prompts and outputs are kept for about 2 days, and metadata is kept longer. Data is not used for training and is not sold.
- **Contact and appeals:** member-service@mail.developer.amd.com.cn